2 зад, 1 конт. а) $L_1 = \underbrace{\{a^n b^{2m} c^k b^{m+2} a^{n+1} / n,m,k \in \mathbb{N}\}}_{L_1'} \cup \underbrace{\{a\}^*\{b\}^*\{a\}^*}_{\text{рег.} \Rightarrow \text{к.св.}}$

Черпова

$S \underset{G}{\to}^{(a)} aSa \underset{G}{\to}^{(2)} a^3Sa^3 \to^{(a)} a^3Baa^3$

$G: S \to aSa \mid Ba$
$B \to bbBb \mid Cbb$
$C \to cC \mid \varepsilon$

$L(G) \stackrel{?}{=} L_1'$

$L_1' \subseteq L(G)$) 1 вариант) индукция по $(n+m+k)$

1.1) $n+m+k=0 \Rightarrow \alpha = bba$ $S \to Ba \to Cbba \to bba$

1.2) ...

2 б.) фикс $n,m,k \in \mathbb{N}$ и $\alpha = a^n b^{2m} c^k b^{m+2} a^{n+1}$

$S \underset{G}{\to}^{(n)} a^nSa^n \underset{G}{\to} a^nBa^{n+1} \underset{G}{\to}^{(m)} a^nb^{2m}Bb^ma^{n+1} \underset{G}{\to} a^nb^{2m}Cb^{m+2}a^{n+1}$

$\downarrow \overset{G}{(k)}$

$a^nb^{2m}\underbrace{c^kb^{m+2}a^{n+1}}_{\alpha} \underset{G}{\leftarrow} a^nb^{2m}c^kCb^{m+2}a^{n+1}$

$\Rightarrow \alpha \in L(G)$

$L(G) \subseteq L_1'$) 1) Ако $\alpha \in L(G^C)$, то $\alpha \in \{c\}^*$ ($\alpha = c^k$ за някое $k \in \mathbb{N}$) (C → сещата с начало C)

2) Ако $\alpha \in L(G^B)$, то $\alpha = b^{2m}c^kb^{m+2}$ за някои $m,k \in \mathbb{N}$

3) Ако $\alpha \in L(G^S) = L(G)$, то $\alpha = a^nb^{2m}c^kb^{m+2}a^{n+1}$, за някои $n,m,k \in \mathbb{N}$



Д-во на 1): Нека $\alpha \in L(G^C) \Rightarrow C \to_G^* \alpha \Rightarrow C \to_G^{(k)} \alpha \Rightarrow C \to_G \alpha_1 \to_G \dots \to_G \alpha_k = \alpha$, $\alpha_0 = C$

Ако $0 \le i < k$, то α_i съдържа един нетерминал и той е $C \Rightarrow \alpha_i$ се получава от α_{i-1} чрез $C \to cC$. $\alpha_k = \alpha \Rightarrow$ не съдържа нет. $\Rightarrow \alpha_k = \alpha$ от α_{k-1} чрез $C \to \varepsilon$.

$\Rightarrow$ за $0 \le i < k$, $\alpha_i = c^i C$

$\Rightarrow \alpha_k = c^{k-1} \in \{c\}^*$

Д-во на 2): Нека $\alpha \in L(G^B) \Rightarrow B \to_G^* \alpha \Rightarrow B \to_G^{(m)} b^{2m} Bb^m \to_G^{(1)} b^{2m} Cb^{m+2} \to_G^{(k)} b^{2m} c^{k-1} b^{m+2}$

$\Rightarrow \alpha = b^{2m} c^k b^{m+2}$ за някои $m, k \in \mathbb{N}$

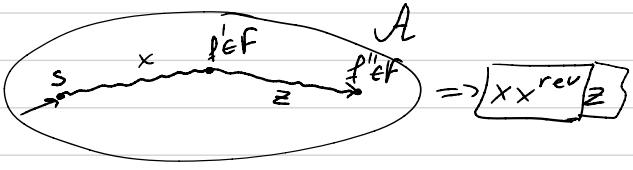
3) е аналогично на 2): $S \to_G^* \alpha \Rightarrow S \to_G^{(n)} a^n Sa^n \to_G^{(1)} a^n Ba^{n+1} \to \dots$

$\Rightarrow \alpha = a^n b^{2m} c^k b^{m+2} a^{n+1}$ за някои $n, m, k \in \mathbb{N}$

5) L-рег

$L' = \{xx^{rev}z \mid x \in L, xz \in L, z \in \Sigma^*\}$ - к. св. ли е?

Нека $A = \langle \Sigma, Q, s, \delta, F \rangle$ е тот. чр-дет. авт. с $L(A) = L$.



$S \to \nu_{p,q'} \nu'_{q',q''}$, $q', q'' \in F$

$\nu_{p,q} \to a \nu_{\delta(p,a),q} a$, $a \in \Sigma$, $p, q \in Q$ $\quad \nu_{p,p} \to \varepsilon, p \in Q$

$N'_{p,q} \to a\, N'_{\delta(p,a),q}$, $a \in \bar{\Sigma}$, $p,q \in Q$

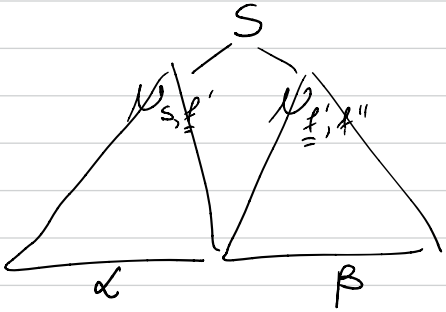
$N'_{p,p} \to \varepsilon$, $p \in Q$

$G = \langle \bar{\Sigma}, N, S, R \rangle$

$N = \{S\} \cup \{N_{p,q} \mid p,q \in Q\} \cup \{N'_{p,q} \mid p,q \in Q\}$

$R = \{S \to N_{s,f'} N'_{f',f''} \mid f', f'' \in F\} \cup \{N_{p,q} \to a\, N_{\delta(p,a),q} \mid a \in \bar{\Sigma}, p,q \in Q\}$

$\cup \{N_{p,p} \to \varepsilon \mid p \in Q\} \cup \{N'_{p,q} \to a\, N'_{\delta(p,a),q} \mid a \in \bar{\Sigma}, p,q \in Q\} \cup \{N'_{p,p} \to \varepsilon \mid p \in Q\}$



1) $\Rightarrow \alpha = xx^{rev}$, за някое x, т.ще $s \overset{x}{\rightsquigarrow}_A f' \in F$

2) $\Rightarrow \beta = z$, за някое z, т.ще $f' \overset{z}{\rightsquigarrow}_A f'' \in F$

$\Downarrow$

$\alpha\beta = xx^{rev}z$ за x, z, т.ще $x \in L$ и $xz \in L$

$\Rightarrow \alpha\beta \in L'$

За $p,q \in Q$: 1) $L(G^{N_{p,q}}) = \{xx^{rev} \mid p \overset{x}{\rightsquigarrow}_A q\}$

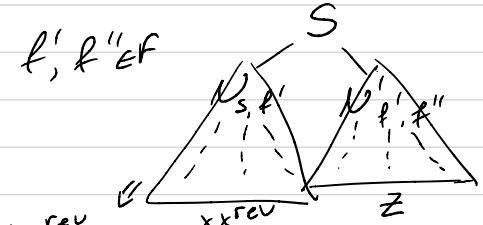
2) $L(G^{N'_{p,q}}) = \{z \mid p \overset{z}{\rightsquigarrow}_A q\}$

$\Rightarrow L(G) \subseteq L'$

Нека $\alpha \in L' \Rightarrow \alpha = xx^{rev}z$ за някои $x, z \in \bar{\Sigma}^*$, $x \in L$, $xz \in L$

$\Rightarrow$ $s \overset{x}{\rightsquigarrow}_A f'$

$s \overset{x}{\rightsquigarrow}_A f' \overset{z}{\rightsquigarrow}_A f''$, за някои $f', f'' \in F$



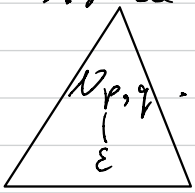
$xx^{rev}z \in L(G)$

α''

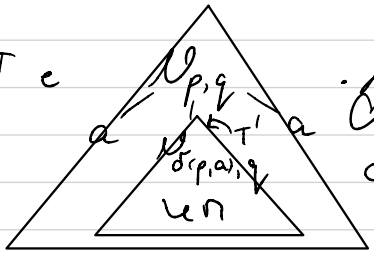
$\Rightarrow L' \subseteq L(G)$

1.1) $\subseteq$) Инд. по височината на дървото за извода

Нека $\alpha \in L(G^{N_{p,q}})$. Нека T е дърво за извод в G, с корен $N_{p,q}$ и етикет α.

1.1.1 сл.) T е 

. Тогава $p=q$ и $\alpha=\varepsilon$. Тогава за $x=\varepsilon$:
$\alpha = xx^{rev}$, $p \overset{x}{\rightsquigarrow}_A q$ (защото $p=q$, $x=\varepsilon$)
$\Rightarrow \alpha \in \{xx^{rev} \mid p \overset{x}{\rightsquigarrow}_A q\}$

1.1.2 сл.) T е 

. Нека α' е етикета на T'.
$\mathcal{O}_T$ (ИП): $\alpha' = xx^{rev}$ за някое x, т.ге $\delta(p,a) \overset{x}{\rightsquigarrow}_A q$

$\alpha = a\alpha' a = axx^{rev}a$

За $x_0 = ax$ имаме $p \overset{a}{\rightsquigarrow}_A \delta(p,a) \overset{x}{\rightsquigarrow}_A q \Rightarrow p \overset{ax=x_0}{\rightsquigarrow}_A q$.

$\alpha = x_0 x_0^{rev} \Rightarrow \alpha \in \ldots$

1.2) $\supseteq$) инд. по $|x|$

1.2.1 сл.) $|x|=0 \Rightarrow x=\varepsilon \Rightarrow xx^{rev}=\varepsilon$

$p \overset{\varepsilon}{\rightsquigarrow}_A q \Rightarrow p=q$

$\Rightarrow N_{p,q} \to_G \varepsilon = xx^{rev}$

$N''_{p,p}$

1.2.2 сл.) $|x|=k+1$ и за вс. $0 \le t \le k$ е вярно, че за вс. $p,q \in Q$ и y, т.ге $|y|=t$, е вярно, че $p \overset{y}{\rightsquigarrow}_A q \to yy^{rev} \in L(G^{N_{p,q}})$

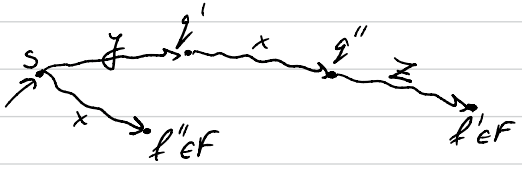
Нека $p,q \in Q$. Нека $p \overset{x}{\rightsquigarrow}_A q$. Нека $x=ay$, $a \in \bar{\Sigma}$, $y \in \bar{\Sigma}^*$, $|y|=k$. $\mathcal{O}_T$ (ИП) $\delta(p,a) \overset{y}{\rightsquigarrow}_A q \to yy^{rev} \in L(G^{N_{\delta(p,a),q}})$

$\Rightarrow p \overset{a}{\rightsquigarrow}_A \delta(p,a) \overset{y}{\rightsquigarrow}_A q \Rightarrow yy^{rev} \in L(G^{N_{\delta(p,a),q}})$

$N_{p,q} \to_G a\, N_{\delta(p,a),q}\, a \to_G^* a\gamma\gamma^{rev}a = xx^{rev} \Rightarrow xx^{rev} \in L(G^{N_{p,q}})$

б) L-рез: $L' = \{ xyx^{rev}z \mid x \in L, yxz \in L \}$ - к.сб. ли е?

Нека $A = \langle \bar{\Sigma}, Q, s, \delta, F \rangle$ е тот. кр. дет. авт. с $L(A) = L$.



$S \to N_{q',q',q'',s}\, N''_{q''}$

$N_{p_y, c_{yx}, t_{yx}, c_x} \to a\, N_{p_y, \delta(c_{yx},a), t_{yx}, \delta(c_x,a)}\, a \mid N'_{s,p_y}$, ако $c_{yx} = t_{yx}$ и $c_x \in F$

$N'_{p,q} \to a\, N'_{\delta(p,a),q} \mid \varepsilon$, ако $p = q$

$N''_p \to a\, N''_{\delta(p,a)} \mid \varepsilon$, ако $p \in F$
$a \in \bar{\Sigma}$